

FLIndex — survey and CPUE data

a4a Training Course — Module 1.4

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Why we need an index at all

Catch data alone cannot tell you whether a small catch means

- **few fish**, or
- **little fishing**

 The identification problem

Catch = $F \times N$. One equation, two unknowns.

A survey gives an independent look at N — that is what breaks the tie.

What an FLIndex holds

Twelve slots — but you normally fill **very few**.

Slot	What it is
index	the abundance series, ages $\times$ years
index.var	variance / weighting, optional
range	ages, years, and survey timing
name, desc	which survey is this
type, distribution	metadata, rarely changed
catch.n, catch.wt, effort, sel.pattern, index.q	usually left empty

For a standard a4a fit: **index plus range**.

The index is relative, not absolute

The survey does not count every fish. It sees a **fraction** of them:

$$I_{a,y} = q_a \cdot N_{a,y}$$

- q is **catchability** — the fraction the survey sees
- The model estimates q ; you only supply I

 Which means

The **units do not matter** and the absolute level does not matter. What matters is how the index changes **through time**.

range — and the two numbers people forget

```
range(idx)[c("startf", "endf")]
```

startf	endf
0.5000000	0.6666667

The survey's start and end as a **fraction of the year** — June to August is 6/12 to 8/12.



Why it matters

They tell the model **how much mortality has already happened** when the survey sees the fish. Set them wrong and the index is compared against the population at the wrong moment.

Age-structured or biomass?

Class	Index is	Use when
FLIndex	numbers at age	your survey is aged
FLIndexBiomass	a single biomass series	no age data, e.g. acoustic

Same idea, same range, same survey timing. FLIndexBiomass simply has one row instead of ten.

One index or many

`sca()` takes `FLIndices` or an `FLIndex`:

```
idx <- FLIndex(idx.flq)      # one survey
```

```
# you can add more surveys
```

```
idxs <- FLIndices(idx1 = idx, idx2 = idx)
```

```
fit <- sca(stock = stk, indices = idxs)
```

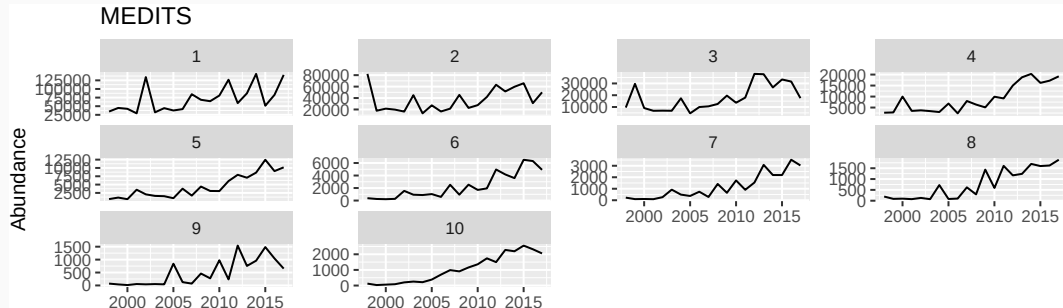


Some methods still require `FLIndices()`

The safest option is to wrap even a single index in an `FLIndices` object.

Looking at it

```
plot(idx) + ggtitle(name(idx))
```



Pitfalls worth knowing

- **Zeros and NAs** are not the same. A true zero is information; a missing survey year should be NA
- **Do not rescale** to match catch — the model estimates q for that
- **One survey per FLIndex.** Two surveys means two objects in FLIndices
- If the survey changed vessel or gear, you need to think if you need to split to two surveys

Summary

- The index is what makes F and N separable
- You normally fill only **index** and **range**
- It is **relative**: trends matter, absolute level does not
- `startf/endif` place the survey inside the year — set them

Next: **Module 1.5 — putting stock and index together, and the first fit.**

Exercise

Open `handson_M1.4_flindex.R` and build the MEDITS index for MUT GSA 32.

1. Read `index.csv` and build the `FLQuant`
2. Create the `FLIndex` and fill the `index` slot
3. Set the survey timing, `name` and `desc`
4. Plot it, and check its dimensions against the stock